NISR 2026 Big Data Hackathon · Track 2

UpliftRwanda: System Demonstration Guide

Unified Socioeconomic Decision-Support Platform for Poverty Reduction & Financial Inclusion

> **Core Objective:** Harmonizing macro-level statistical surveys with actionable local interventions across Rwanda's 30 districts to support NST2 and Vision 2050 targets.

1. System Architecture & Multi-Survey Ingestion

UpliftRwanda breaks down traditional data silos by ingesting and harmonizing **1,020 calibrated data points** across five major national survey vintages into a secure PostgreSQL backend served via FastAPI:

- **FinScope 2024:** Financial inclusion (96%), savings, and credit pyramids.
- **EICV7:** Household living conditions and VUP poverty headcounts (~41%).
- **CFSVA 2024:** Food security, dietary diversity (HDDS), and child stunting.
- **RDHS 2025:** Living standards, wealth quintiles (hv270), and child cohort mortality.
- **LFS 2025:** Labour force participation, unemployment (12.4%), and youth NEET (24.5%).

2. Core Functional Modules

Module A: Labour Intelligence

Tracks district employment friction, youth inactivity clusters, and gender gaps (14.3 pp LFPR differential).

Module B: VUP Social Protection

Monitors 671.8k beneficiaries across 5 pillars, tracking public works timelines and SACCO disbursement rails.

3. Interactive Features & Matchmaker Engine

The platform translates macro indicators into grassroot utility via two signature interactive tools:

- **Inclusion & Opportunity Matchmaker (`POST /api/matchmaker/recommendations`):** An explainable rule-based engine matching individual user profiles (age, gender, occupation, district, savings goal) against a 12-product financial stack (Ejo Heza, Umurenge SACCO, VUP tracks, BDF).
- **Policy Scenario Lab:** A ridge-blended OLS elasticity model allowing leaders to simulate 1-to-5-year outcomes by adjusting mobile money and credit levers.

4. Boardroom Executive Brief Generator

Eliminates hours of manual report compilation by streaming professional, Cabinet-grade PDF briefs (`GET /api/brief/{district}.pdf`) using ReportLab, combining survey evidence, automated policy verdicts, and targeted interventions.

5. Live Presentation Pitch Script (3 Minutes)

> **0:00 – 0:45 (The Hook):** Introduce UpliftRwanda as the unification of 1,020 data points across 5 survey vintages.
> **0:45 – 1:15 (Spatial & District Deep-Dive):** Click through the district heatmap (e.g., Kicukiro or Burera) to demonstrate instant data synchronization.
> **1:15 – 2:00 (Multi-Sectoral Modules):** Showcase Labour LFS 2025, VUP Tracker, and the FinScope Matchmaker.
> **2:00 – 2:30 (Predictive Scenario Lab):** Demonstrate policy-lever simulations on poverty trajectories.
> **2:30 – 3:00 (Boardroom Briefs):** Generate a one-click PDF brief and conclude.